7.1 Adding and Subtracting Polynomials

Name _____ Period ____ Date _____

Algebra I 7.1 · Student sheet

I can…

- find the degree of a monomial.
- write a polynomial in standard form and classify it by its degree and number of terms.
- add and subtract polynomials, including ones with two variables.

Vocabulary · fill in the missing word

Monomial — a number, a variable, or a product of a number and variables with _____ exponents

Degree of a monomial — the sum of the _____ of its variables

Polynomial — a monomial or a sum of _____

Binomial / trinomial — a _____ with two / three terms

Degree of a polynomial — the _____ degree of its terms

Standard form — terms in order of _____ exponents

Leading coefficient — the coefficient of the first term in _____ form

Part 1 · Monomial sort

Sort the cards. Write each expression in the correct column.

Monomial	Not a monomial

An expression is a monomial when _____

Part 2 · Notes: fill in the blanks

Monomial: a number, a variable, or a _____ of a number and variables with _____ exponents.

Degree of a monomial: the _____ of the exponents of its variables.

Polynomial: a monomial or a _____ of monomials. Binomial: ____ terms. Trinomial: ____ terms.

Degree of a polynomial: the _____ degree of its terms.

Standard form: exponents _____ from left to right. **Leading coefficient:** the coefficient of the _____ term.

To subtract a polynomial, add its _____: change _____ sign.

Part 3 · Examples (work them with the class)

Example 1

Find the degree of each monomial.

a. $\frac{1}{2}x^3$ b. $-3xy^4$ c. 8 d. $4x^2yz^3$

Example 2

Write the polynomial in standard form. Identify the degree and leading coefficient. Then classify it by the number of terms.

$15x - x^3 + 3$

Example 3

Find the sum.

$(2x^3 - 5x^2 + x) + (2x^2 + x^3 - 1)$

Example 4

Find the difference.

$(4n^2 + 5) - (-2n^2 + 2n - 4)$

Example 5

Find the difference.

$(5a^2 - 3ab + 2b^2) - (a^2 + ab - 4b^2)$

Example 6

A store sells x shirts. The revenue is $R = 18x$ and the cost is $C = 0.02x^2 + 6x + 150$ (in dollars). Write a polynomial for the profit $P = R - C$. Then find the profit when the store sells 200 shirts.

Part 4 · Partner practice

Partner 1

Find the sum.

$(3y^2 - 7y + 4) + (-5y^2 + y + 9)$

Partner 2

Find the difference.

$(6k^3 + k - 2) - (2k^3 - 4k^2 + k - 8)$